Similarity Error Analysis

Find the mistake. Explain it. Fix it.

Name: _____ Period: _____ Date: _____

1. Setting up a proportion

ORIGINAL PROBLEM

△ABC ~ △DEF with AB = 6, DE = 9 and BC = 8. Find EF.

INCORRECT WORKED SOLUTION

$$\frac{AB}{DE} = \frac{EF}{BC}$$

$$\frac{6}{9} = \frac{x}{8}$$

$$9x = 48$$

$$x = \frac{16}{3}$$

A. EXPLAIN THE MISTAKE (1 POINT)

Describe exactly which step is wrong and why.

B. SHOW CORRECT WORK (1 POINT)

2. Area and scale factor

ORIGINAL PROBLEM

Two similar figures have scale factor 3. The smaller has area 10. Find the larger area.

INCORRECT WORKED SOLUTION

Area = 10 · 3

Area = 30

A. EXPLAIN THE MISTAKE (1 POINT)

Describe exactly which step is wrong and why.

B. SHOW CORRECT WORK (1 POINT)

Scoring: Explanation = 1 point • Correct work/answer = 1 point • Total = 20 points

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3. Perimeters of similar figures

ORIGINAL PROBLEM

Similar polygons have scale factor 2 : 5. The smaller has perimeter 24. Find the larger perimeter.

INCORRECT WORKED SOLUTION

$\frac{2}{5} = \frac{x}{24}$

$5x = 48$

$x = 9.6$

A. EXPLAIN THE MISTAKE (1 POINT)

Describe exactly which step is wrong and why.

B. SHOW CORRECT WORK (1 POINT)

4. AA similarity

ORIGINAL PROBLEM

△ABC has angles of 50° and 60°. △DEF has angles of 60° and 80°. Are they similar?

INCORRECT WORKED SOLUTION

Both triangles have a 60° angle.

AA similarity applies.

The triangles are similar.

A. EXPLAIN THE MISTAKE (1 POINT)

Describe exactly which step is wrong and why.

B. SHOW CORRECT WORK (1 POINT)

Scoring: Explanation = 1 point • Correct work/answer = 1 point • Total = 20 points

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5. Indirect measurement

ORIGINAL PROBLEM

A 6-ft person casts a 4-ft shadow. At the same time, a tree casts a 20-ft shadow. Find the tree's height.

INCORRECT WORKED SOLUTION

$$\frac{6}{4} = \frac{20}{x}$$

$$6x = 80$$

$$x = \frac{40}{3} \approx 13.3 \text{ ft}$$

A. EXPLAIN THE MISTAKE (1 POINT)

Describe exactly which step is wrong and why.

B. SHOW CORRECT WORK (1 POINT)

6. Side-splitter theorem

ORIGINAL PROBLEM

In $\triangle ABC$, $DE \parallel BC$ with D on AB and E on AC. $AD = 4$, $DB = 6$ and $AE = 6$. Find EC.

INCORRECT WORKED SOLUTION

$$\frac{AD}{AB} = \frac{AE}{EC}$$

$$\frac{4}{10} = \frac{6}{x}$$

$$4x = 60$$

$$x = 15$$

A. EXPLAIN THE MISTAKE (1 POINT)

Describe exactly which step is wrong and why.

B. SHOW CORRECT WORK (1 POINT)

Scoring: Explanation = 1 point • Correct work/answer = 1 point • Total = 20 points

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7. Volume and scale factor

ORIGINAL PROBLEM

Two similar prisms have scale factor 2. The smaller has volume 12. Find the larger volume.

INCORRECT WORKED SOLUTION

Volume = 12 · 2

Volume = 24

A. EXPLAIN THE MISTAKE (1 POINT)

Describe exactly which step is wrong and why.

B. SHOW CORRECT WORK (1 POINT)

8. Finding a scale factor

ORIGINAL PROBLEM

A side of length 5 corresponds to a side of length 15 in a similar figure. Find the scale factor from small to large.

INCORRECT WORKED SOLUTION

Scale factor = 15 − 5

Scale factor = 10

A. EXPLAIN THE MISTAKE (1 POINT)

Describe exactly which step is wrong and why.

B. SHOW CORRECT WORK (1 POINT)

Scoring: Explanation = 1 point • Correct work/answer = 1 point • Total = 20 points

Similarity Error Analysis

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Name: _____ Period: _____ Date: _____

9. Differences vs. ratios

ORIGINAL PROBLEM

Are triangles with sides 3, 4, 5 and 6, 7, 8 similar?

INCORRECT WORKED SOLUTION

Each side is 3 more than the matching side.

6 = 3 + 3, 7 = 4 + 3, 8 = 5 + 3

The triangles are similar.

A. EXPLAIN THE MISTAKE (1 POINT)

Describe exactly which step is wrong and why.

B. SHOW CORRECT WORK (1 POINT)

10. Similar rectangles

ORIGINAL PROBLEM

Are a 4 by 6 rectangle and a 6 by 8 rectangle similar?

INCORRECT WORKED SOLUTION

Both are rectangles, so all angles are 90°.

The angles match, so the rectangles are similar.

A. EXPLAIN THE MISTAKE (1 POINT)

Describe exactly which step is wrong and why.

B. SHOW CORRECT WORK (1 POINT)

Scoring: Explanation = 1 point • Correct work/answer = 1 point • Total = 20 points